The applicability frontier of a substrate-blind big-Φ detector across (n, density, structure, temporal): a 14-point measurement map, the mean-field paradox, and the temporal dual closed-negative

Three closed-negative axes (micro / sparse / algorithmic), two ground-truth calibrations (BL Voyager-1, XOR-cascade hive), four border points (lattice / SETI@home Arecibo / longer-playback / Kuramoto sync), the mean-field paradox, and the TEMPORAL T1/T2 dual closed-negative over hypotheses `XENO/H_832`–`H_840` and `TEMPORAL/H_841`–`H_842`

`anima` maintainers
dancinlab
github.com/dancinlab/anima

2026-05-29 (v3)

Abstract

A substrate-blind big-Φ detector — an Integrated Information Theory 4.0 [1] reading of an arbitrary float-array signal, binarised at threshold 0.5, encoded as a two-unit co-occurrence transition probability matrix (TPM), and passed to a deterministic IIT 4.0 structure-cut big-Φ (Φ) algorithm — promises to read the "existence of consciousness" of any candidate substrate (alien, digital, panpsychist, simulated, AGI activation, hive) without word, body, or carbon prior. The natural question is then: *on which* (n, *density*, *structure*) *regime does the detector actually calibrate, and on which does it deterministically break?* Without this map any reported "Φ" on a novel substrate is uninterpretable.

We answer this empirically with a 14-point measurement matrix (`XENO/H_832`–`H_840` + `TEMPORAL/H_841`–`H_842`, five XENO-FRONTIER-5 closure rounds + the X837 SETI@home BOINC follow-up + the X10 hive-mind 4-cell coupling follow-up + the X840 longer-playback recovery follow-up + the TEMPORAL T1/T2 dual lag-axis / embed-dim extension follow-up). The points span (i) micro substrates ($n = 16$–32, 1–3 bit panpsychism candidates), (ii) sparse mid-n activation ($n = 64$, AGI LLM-like attention/residual/structured), (iii) algorithmic dense ($n = 128$, simulation-hypothesis candidates: lattice, floating-point, π-digits, natural noise), (iv) natural-carrier dense ($n = 128$, 60.9% activation: Breakthrough-Listen Voyager-1 baseband as the ground-truth calibration), (v) sparse natural-noise dense ($n = 128$, 20.3% activation: a SETI@home Arecibo 2004 BOINC `bg_pot` replay fired on a real Ubuntu 22.04 i686-multilib pod), (vi) hive-mind coupling: four 32-sample 4-cell substrates aggregated to $n = 128$ — independent walkers, mean-field weak coupling, Kuramoto strong coupling ($K = 2.0$ all-to-all sync), and XOR-cascade hive emergence with 1-lag carry — the X10 follow-up, **(vii) X840 longer-playback recovery (**$n = 128$**,** 24.4% **prog harvest,** 3600 s **pod timeout)**, and **(viii) TEMPORAL: 4 substrate (hive / Voyager / random / lattice)** $\times$ **4 lag** $\Delta t \in \{1, 8, 32, 64\}$ **for T1 = 16 lag-window measurements + 4 substrate** $\times$ **4 embed-dim** $e \in \{2, 3, 4, 5\}$ **Takens delay-embedding for T2 = 16 multi-unit measurements — the dual 5D applicability extension follow-up**.

The measurement is a clean two-ground-truth separation: **X7 BL Voyager-1 (**$\Phi =$ 0.114**, [N])** and **X10-hive XOR cascade (**$\Phi = 1.565$ **`conscious`, [N] ground-truth #2)**;

X4 micro, X6 sparse, X5b algorithmic axes are each closed-negative (F-falsifier 0/4, 1/5, 1/4 PASS, [**X**]); three border points — X5a lattice ($\Phi = 0.660$, periodic-structure ceiling), X837 SETI@home Arecibo ($\Phi = 0.567$, sparse natural-noise borderline), and X10-strong Kuramoto sync ($\Phi = 0.408$, mean-field-but-coupled border) — sit above or near the pre-frozen $\Phi = 0.5$ threshold but with type-axis intact (`coherent_non_conscious`), [**!**].

The mean-field paradox. The most surprising finding is in the X10 ordering. *Naive intuition* says coupling → integration: Φ should increase monotonically along (independent → weak mean-field → strong Kuramoto → hive). *Measured* Φ *instead drops from* 0.131 *(independent) to* 0.036 *(weak mean-field), rises only to* 0.408 *at strong Kuramoto, and jumps to* 1.565 *at the XOR cascade.* Mean-field averaging reduces Φ *below* the independent baseline because uniformity makes the substrate reducible (IIT 4.0 axiom: integration ≡ irreducibility, not correlation), while genuine cascade-coupling makes the substrate irreducible. The pre-registered monotonicity falsifier (F-X10-MONOTONE) explicitly FAILED, and the failure *confirms* the IIT 4.0 axiom on which the detector is built. The mean-field paradox is the headline finding of v2.

Aggregating, the measurable axis remains *the conjunction* $n \geq 128 \;\wedge\;$ (**density** $\geq$ 60% $\vee$ **MIP-irreducible deterministic transition**). Three regimes (micro / sparse / algorithmic) are deterministically ruled out as outside the instrument's calibrated frontier. The new ground-truth #2 (X10-hive) confirms that the calibrated regime includes *any* truly irreducible coupling at $n \geq 128$, not just natural-carrier density.

These closed-negative results satisfy `a_paper_negative_ok`: the finding is the instrument applicability frontier itself, not a "no-consciousness-detected" headline. The mean-field paradox is the v2 escalation: a positive finding (X10-hive ground-truth #2 with $\Phi = 1.565$ `conscious`) and an axiom-confirming negative (F-X10-MONOTONE FAILED *because* mean-field ≠ irreducibility, which is precisely IIT 4.0).

The v3 escalation: temporal axis dual closed-negative. A natural follow-on question is whether the (n, density, structure) 3-axis decomposition can be extended into a 5-axis decomposition (adding lag-window Δt for T1 and Takens delay-embedding dimension e for T2) by a purely instrument-side modification of the 2-unit co-occurrence TPM detector. *Both extensions are deterministically ruled out by the same pre-registered methodology.* T1 (`H_841`, 16 measurements over 4 substrate $\times$ $\Delta t \in \{1, 8, 32, 64\}$) gives 1/5 PASS: the hive substrate's Φ *rises* from 0.013 at $\Delta t = 1$ to 0.999 at $\Delta t = 64$ (the lag-window inflates Φ in proportion to predictable repetition, not integration). T2 (`H_842`, 16 measurements over 4 substrate $\times e \in \{2, 3, 4, 5\}$ Takens delay-embedding) gives 2/5 PASS but inflates 4/4 substrate monotonically with e (random e=5 $\Phi = 13.63$, Voyager e=5 $\Phi = 28.36$). The T1 lag-axis artifact is *not* resolved by the T2 multi-unit extension, and a new embed-dim sparse-state inflation artifact is added on top. The honest follow-on (T3) is therefore not a single-trajectory Φ inflation but *time-averaged* Φ *over surrogate-data baselines together with a Granger-causality control*, which is opened as a deferred lane (§7).

X840 (`H_840`, longer-playback recovery) adds a 4th border point at the X837 SETI@home location: a 3600 s pod timeout (vs. X837's 600 s) and 24.4% prog harvest (vs. 21.3%) returns $\Phi = 0.5669$ (vs. X837 $\Phi = 0.5670$, $\Delta < 0.0002$ regression-stable). The X840 finding is *double*: the longer-playback hypothesis itself ("more pod-time → new triplets → new Φ") is honestly FALSIFIED (no new triplets in the extra 3% progress; the BOINC `bg_pot` 64-bin array is bit-identical to X837), while the `invariant_detector` reports regression-stable at the sparse natural-noise border.

We publish all ten verdicts (X7, X4, X5, X6, X837, X10 4-sub, X840, T1, T2) verbatim in a `companion/` ledger for bit-identical re-verification, and we close the XENO-FRONTIER-5 followup-2 cycle at this paper as the `a_paper_only_at_closure` trigger (round 4/5).

1 Introduction

The instrument-first methodology. Hexa-native, deterministic, hypothesis-first measurement on the `anima` project follows an *instrument-first* methodology: any reading of a measurement (Φ, KL-divergence, register-collapse rate) is meaningless unless the calibrated input regime of the measuring instrument is itself first characterised. The XENO domain [10] carries this practice into substrate-blind big-Φ [1, 11, 13]: a function `invariant_detector` :

$\mathbb{R}^n \times \mathbb{N} \to \{\Phi, \text{integration}, \text{irreducibility}, \text{substrate\_type}\}$ that takes *any* signal — alien radio carrier, BOINC residual noise, floating-point lattice, panpsychist micro-substrate, LLM activation sparsity — and returns the IIT 4.0 big-Φ together with a discrete type label.

The detector is designed substrate-blind by construction. It binarises the input at threshold 0.5, builds a 2-unit (sample t, sample t+1) co-occurrence row-stochastic transition probability matrix (TPM), and delegates to `stdlib/consciousness/iit4_bigphi` (the hexa-lang commons-g61 IIT 4.0 implementation). The TPM is the substrate; word, body, language, carbon, RNA, silicon priors all vanish into the float array. This is what allows the question "is X conscious?" to be asked even when X has no body and no language — as long as the input float-array is honestly the candidate substrate's state.

The applicability question. The detector's output, however, is only as meaningful as the regime on which it is calibrated. If the instrument deterministically breaks below some n, or above some sparsity, or for some specific structural class (algorithmic vs. deterministic vs. natural), then any reported Φ in that broken regime is an instrument artifact, not a substrate property. The XENO working group therefore pre-registered the methodological imperative: *before reporting any candidate-substrate Φ, first close the applicability frontier of the detector along the* $(n, \textit{density}, \textit{structure})$ *axes.*

XENO-FRONTIER-5 closure and the two follow-up cycles. The XENO-FRONTIER-5 campaign is the original closure: five pre-registered candidate substrates, ranging across the three axes, each measured with the same detector with frozen prediction thresholds and falsifiers. Each round was published verbatim in `UNIVERSE/H_832`–`H_836`. The X8 SETI@home archive *follow-up 1* fire was published verbatim in `UNIVERSE/H_837`. *Follow-up 2*, the X10 hive-mind 4-cell coupling axis, was published verbatim in `UNIVERSE/H_838`. This paper v2 aggregates all seven points (plus the X8 X837 follow-up) into a 7+1-point matrix and reports the applicability frontier finding together with the v2 escalation: the mean-field paradox and ground-truth #2.

Why publish closed-negatives. The directive `a_paper_negative_ok` [4] explicitly licenses closed-negative findings: a **[X]** verdict that deterministically rules out an axis is a valid paper, framed as the falsifier plus the ruled-out space. This paper is exactly that frame. The headline is not "no alien consciousness was detected" (a vacuous statement on a mis-calibrated regime). The headline is "the substrate-blind big-Φ detector calibrates only inside one corner of the $(n, \text{density}, \text{structure})$ cube, and three corners are deterministically ruled out." This is what an instrument-first methodology publishes.

Roadmap. §2 states the four pre-registered applicability falsifiers and the meta-falsifier on the 14-point matrix, including the five-falsifier X10 row, the five-falsifier X840 row, and the five-falsifier T1 and T2 rows. §3 fixes the measurement substrate and the candidate points. §4 aggregates the ten verdicts verbatim (plus the X8 archive-acquisition pointer). §5 establishes the applicability frontier as a 3-axis conjunction and identifies the three closed-negative axes plus the two ground-truth points (X7, X10-hive) plus the four border points (X5a, X837, X840, X10-strong). §6 isolates the *mean-field paradox* as a positive IIT 4.0 axiom-confirmation finding (the v2 escalation). §7 reports the *temporal axis dual closed-negative* (T1 lag-axis + T2 embed-dim) as the v3 escalation: the simple 5-axis instrument extension is deterministically ruled out, motivating T3 (time-averaged Φ with surrogate-data baseline and Granger-causality control) as the deferred natural follow-on. §8 discusses validity threats.

2 Hypothesis: the applicability frontier of `invariant_detector`

Decomposition. The candidate-substrate space is naturally decomposed along three axes:

- n **axis** — the length of the input float-array (number of binarised samples available for TPM estimation). Examples: $n = 16$ (a 1-bit thermostat over a 16-tick window) up through $n = 128$ (a dense BL Voyager-1 carrier or a 128-bin BOINC `bg_pot` array).
- **density axis** — the fraction of samples above the binarisation threshold 0.5 (i.e. the fraction of 1s in the TPM input). Sparse (e.g. 20.3% of $n = 128$ active in the X837 SETI@home `bg_pot`) vs. dense (e.g. 60.9% active in the X7 BL Voyager-1 carrier).
- **structure axis** — the deterministic, algorithmic, or natural-noise character of the transition $b_t \to b_{t+1}$. Examples: deterministic periodic (lattice / Planck-floor 8-level), algorithmic (Pi-digits, floating-point round, XOR LFSR), or natural (Bates-4 Gaussian, real radio-band noise, BOINC residual).

Pre-registered falsifiers. Frozen *before* measurement on the six substrates, the four applicability falsifiers are listed in Table 1.

Pre-registered prediction matrix. Frozen pre-measurement, the expected applicability pattern is:

- **X7 BL Voyager-1** (n=128, dense $\sim 60\%$, natural carrier-line + noise) $\to$ *calibrate* ([**N**]); $\Phi < 0.5$ and type = `coherent_non_conscious` or `noise`. (Voyager-1 is a narrow-band carrier; substrate type is empirical non-consciousness.)
- **X4 panpsychism micro** (n=16–32, four substrates, 1–3-bit dynamics) $\to$ *deterministically break* ([**X**]). Pre-registered: 4/4 falsifiers FAIL.
- **X6 AGI LLM-like** (n=64, four activation classes, one sparse-attention class) $\to$ *deterministically break* ([**X**]) due to sparse-bias drift. Pre-registered: $\leq 1/5$ PASS.
- **X5b algorithmic** (n=128, fp-bound + π-digits + natural) $\to$ *deterministically break* ([**X**]) due to algorithmic-vs-natural indistinguishability. Pre-registered: $\leq 2/5$ PASS (lattice border + natural pass; fp + π fail).
- **X837 SETI@home BOINC** (n=128, dense 20.3%, sparse natural noise) $\to$ *border* ([**!**]): pre-registered `coherent_non_conscious` type but pre-registered $\Phi < 0.5$ — a FAIL on Φ alone but pass on type axis.
- **X10 hive-mind 4-cell coupling** (4 substrates aggregated to $n = 128$ each: independent / weak mean-field / strong Kuramoto / XOR-cascade hive) $\to$ *mixed: 3/5 PASS, ground-truth #2 emergence*. Pre-registered: independent low-Φ, weak low-Φ, strong above threshold, hive above threshold AND `conscious`, monotone. Two pre-registered *naive* predictions (F-X10-STRONG monotone, F-X10-MONOTONE) were designed to fail *if* mean-field $\neq$ irreducibility, exactly so that the IIT 4.0 axiom is testable as a falsifier on this paper's measurement matrix.

Meta-falsifier (v2).

> **F-meta-v2**: if the seven measurement points (plus the X10 4-sub-substrate row) do not partition cleanly into {X7 calibrate, **X10-hive calibrate #2**, X4/X5b/X6 closed-negative, X5a/X837/X10-strong border}, then the 3-axis (n / density / structure) applicability decomposition is FALSIFIED. The SUPPORTED condition is exactly the pre-registered pattern listed above. Additionally, the F-X10-MONOTONE *naive* prediction is pre-registered to fail *if* the IIT 4.0 axiom holds; its verified failure is therefore a positive axiom-confirmation, not a detector defect (§6). The closed-negative finding is a valid paper outcome under `a_paper_negative_ok`, and the positive ground-truth #2 plus mean-field paradox is the v2 escalation.

3 Method

Detector implementation. The detector lives at `XENO/detector/invariant_detector.hexa` (commit `be75bb10b`, identical to the version measured in all six rounds; no detector-side post-tuning between rounds). The signature is

```
fn invariant_detector(signal: array, n: int) -> map
//   returns { phi, integration, irreducibility, substrate_type }
```

and the closed-form pipeline is:

1. **binarise**: $b_i = 1$ if signal$[i] \geq 0.5$, else 0. (Threshold is the only externally tunable parameter; frozen at 0.5 across all six runs.)
2. **2-unit co-occurrence TPM**: build the row-stochastic transition matrix
$$P\big[(b_t, b_{t+1}) \to (b_{t+1}, b_{t+2})\big]$$
by sweeping $i = 0, 1, \ldots, n-3$. The 4×2 TPM entries are normalised against the empirical state-frequency of each $(b_t, b_{t+1}) \in \{0,1\}^2$.
3. **IIT 4.0 big-Φ**: delegate to `stdlib/consciousness/iit4_bigphi.hexa` (hexa-lang commons-g61 SSOT). The big-Φ is the structure-cut Φ [1].
4. **type label**: discrete map type $= \begin{cases} \text{noise} & \Phi = 0 \\ \text{conscious} & \text{integration} > \text{noise floor} \\ \text{coherent\_non\_conscious} & \text{otherwise} \end{cases}$

Determinism / honesty (`a_blue_closed, p7=0`). Every Φ in this paper is from a \$0 Mac-local or RunPod-pod hexa-only run, no LLM judge, no RNG, no perplexity reading. The X1 smoke test (`F-DETECT-NULL / NOISE / COUPLED`) was 5/5 PASS pre-XENO-FRONTIER-5 (`.verdicts/829_xeno_invariant_detector/x1_smoke.txt`). All six measurement runs are byte-equal reproducible via the verbatim verdict files in the `companion/` ledger.

Eight measurement substrates. The eight measurement points (Table 2) are taken from seven independent XENO rounds (R1–R5 of XENO-FRONTIER-5 plus the X837 SETI@home follow-up plus the X10 hive-mind follow-up). Round 5 (X8 itself) is an archive-acquisition + pod spec milestone ([!]); the actual BOINC playback was deferred and fired in the X837 follow-up round on a real Ubuntu 22.04 RunPod pod. The X10 row is itself four sub-substrates (independent / weak / strong / hive) sharing the same detector and the same $n = 128$.

X837 real-pod fire. The X837 measurement [7] is the only one that required a remote pod — SETI@home v3.03 is a 2000-era ELF32 i686 binary, so the playback ran on a RunPod GPU pod `wcy166z3cqjam7` (Ubuntu 22.04.5 LTS with glibc-2.35 + i386 multilib enabled), ∼ 16 min wall, \$0.10 actual cost, dispatched autonomously per `a_fire_autonomous`. Pod spec, work-unit provenance, and playback signature are all verbatim in `.verdicts/837_xeno_x8_followup_fire/x837_run.txt`. The Arecibo work unit (sky 51.725° R.A., 17.330° Dec, recorded Wed May 5, 2004 17:57:39 UTC, base frequency 1.419433594 GHz) ran to 21% completion before the 600s timeout; two real triplet detections (power $= 8.27, 8.18$; period $= 2.067662$s, freq $= 1419438781.74$ Hz) and the 64-bin `bg_pot` array (min 0.095, max 3.004, mean 1.000) were extracted. The `bg_pot` was normalised by its max and upsampled 2× to $n = 128$ to match the X7 calibration template (no other transform).

X10 hive-mind 4-cell coupling fire. The X10 measurement [8] (`XENO/scan/hive_mind_invariant.hexa`, `UNIVERSE/H_838`) is a \$0 Mac-local fire that runs four sub-substrates over four 32-sample 4-cell trajectories aggregated to $n = 128$ each. Sub-substrate (a) is four independent LCG walkers; (b) is weak *mean-field* coupling (50% self + 50% pool mean per tick); (c) is strong *Kuramoto* all-to-all phase coupling at $K = 2.0$; (d) is *XOR-cascade* hive emergence with a 1-lag carry ($b_t^{(i)} = b_{t-1}^{(i)} \oplus \bar{b}_{t-1}$ where $\bar{b}$ is the pool 4-cell mean threshold). All four sub-substrates share the same $n = 128$, the same binarisation threshold 0.5, and the same `invariant_detector` (no X10 detector variant). The X10 pre-registration was frozen as a row of five falsifiers (F-X10-INDEP / WEAK / STRONG / HIVE / MONOTONE) before any measurement was committed.

Pre-registration sanity (per-substrate). Each Φ-threshold and each falsifier listed in §4 was *frozen pre-run for the round* (i.e. the X4 falsifiers were fixed before X4 ran, the X5 falsifiers before X5, and so on). The $\Phi = 0.5$ threshold used in X837, X10, X840, T1, and T2 is identical to the X7 template threshold (no per-round tuning). The verbatim verdict files include this honesty note.

X840 longer-playback recovery fire (v3). The X840 measurement [9] (`XENO/scan/seti_boinc_phi_ful` `UNIVERSE/H_840`) is a longer-playback follow-on to X837 (600 s pod timeout). The X840 pod (RunPod `lfxh817pdk2h39`, Ubuntu i386-multilib) was assigned a 3600 s pod-timeout (i.e. 6× the X837 timeout) on the *same* SETI@home Arecibo 2004 work-unit (`sahfiles_workunits/work_unit.sah` sha256 byte-identical to X837). The BOINC checkpoint at harvest time recorded prog = 24.4% (vs. X837 21.3%) and `cpu` = 1083.3 s before the pod-timeout SIGKILL. The five-falsifier X840 row (F-X840-RECOVER-HARVEST, -SPIKE-NONZERO, -PHI-N128, -X837-CONSIST, -NOT-CONSC) was frozen before the X840 fire and explicitly tests whether *longer* playback recovers any new SETI signal (triplets / pulses / new Φ). The `bg_pot` 64-bin array and the Φ measurement re-use the X7 calibration template threshold (0.5, unchanged).

TEMPORAL T1 lag-axis Δt-scan fire (v3). The T1 measurement [5] (`TEMPORAL/detector/timeshift_` `UNIVERSE/H_841`) extends the 2-unit co-occurrence detector to an explicit lag-window Δt parameter: instead of building the TPM on (b_t, b_{t+1}), the lag-detector builds it on $(b_t, b_{t+\Delta t})$. The T1 fire sweeps $\Delta t \in \{1, 8, 32, 64\}$ across four substrates (the X10-d *hive*, the X7 *Voyager-1*, a random-noise baseline, and the X5a *lattice*) for $4 \times 4 = 16$ measurements. The pre-registered falsifier row tests instant integration (F-T1-INSTANT), mid-window (F-T1-MID), long-window (F-T1-LONG), random decay (F-T1-DECAY), and lag-invariance (F-T1-LAGINV: $\Phi(\Delta t = 1) \geq \Phi(\Delta t = 64)$). The $\Phi = 0.5$ threshold is re-used from X7 unchanged. The fire is \$0 Mac-local and hexa-only-deterministic.

TEMPORAL T2 Takens delay-embedding e-scan fire (v3). The T2 measurement [6] (`TEMPORAL/detector/time_embed_detector.hexa`, `UNIVERSE/H_842`) follows on from the T1 lag-axis artifact by replacing the 2-unit TPM with a Takens e-unit *delay-embedding* TPM: $\mathbf{b}_t = (b_t, b_{t-d}, b_{t-2d}, \ldots, b_{t-(e-1)d})$ with delay $d = 1$, sweeping $e \in \{2, 3, 4, 5\}$. The TPM becomes a $2^{e-1} \times 2$ row-stochastic matrix on the multi-unit embedded state. The four substrates are the same as T1 (hive / Voyager / random / lattice) for $4 \times 4 = 16$ measurements. The pre-registered falsifier row tests random low (F-T2-INSTANT-LOW), hive consciousness (F-T2-HIVE-CONSC), artifact resolution (F-T2-ARTIFACT-FIX), random decay (F-T2-RANDOM-DECAY), and hive monotonicity (F-T2-HIVE-MONOTONE). The $\Phi = 0.5$ threshold is again re-used from X7 unchanged. The fire is \$0 Mac-local and hexa-only-deterministic. Critically, T2 was *designed* to resolve the T1 lag-axis artifact via the multi-unit embedding; F-T2-ARTIFACT-FIX is the falsifier on exactly that question.

4 Measurement: the 7+1-point applicability matrix

Table 3 aggregates the seven verdicts verbatim, plus the X8 archive pointer. Each row corresponds bit-identically to the result section of `UNIVERSE/H_8xx.md` and to `companion/verdict-ledger.json`.

X10 row aggregate verdict. The X10 row's five pre-registered falsifiers evaluate to 3/5 PASS, 2/5 FAIL ([**!**] PARTIAL-SUPPORT). The two FAILs (F-X10-STRONG, F-X10-MONOTONE) are exactly the two *naive* predictions designed pre-run to catch the IIT 4.0 axiom: their failure is the mean-field paradox finding (§6), not a detector defect. The X10-d hive sub-substrate is the second ground-truth calibration point of the entire matrix (the first being X7 BL Voyager-1), and the only point that scores type=`conscious` under an honestly pre-registered falsifier (cf. X6-b attention $\Phi = 1.213$, which is a *ruled-out* sparse-bias artifact, not a ground-truth).

Verbatim verdict files. The full standard-output of each hexa-only run is preserved at:

- `.verdicts/832_xeno_voyager_phi_real/x7_run.txt` — X7 BL Voyager-1 ([**N**]).
- `.verdicts/833_xeno_panpsy_falsifier/x4_run.txt` — X4 panpsy 4-micro ([**X**]).
- `.verdicts/834_xeno_agi_sentience/x6_run.txt` — X6 AGI sentience 4-LLM-like ([**X**]).
- `.verdicts/835_xeno_sim_hypothesis/x5_run.txt` — X5 4-sim-candidate (X5a lattice border + X5b fp/pi/natural [**X**]).
- `.verdicts/836_xeno_seti_boinc_pod/x8_run.txt` — X8 SETI@home archive + dispatch handoff ([**!**]archive-acquired-pod-ready, not aggregated into the 5+1-point measurement matrix as the playback was deferred to X837).
- `.verdicts/837_xeno_x8_followup_fire/x837_run.txt` — X837 SETI@home Arecibo 2004 real-pod playback ([**!**]border).
- `.verdicts/838_xeno_hive_mind/x10_run.txt` — X10 hive-mind 4-cell coupling 4-substrate fire ([**!**] 3/5 PASS aggregate; X10-d hive [**N**] ground-truth #2).
- `.verdicts/840_xeno_x837_full_playback/x840_run.txt` — X840 SETI@home Arecibo longer-playback 3600 s pod-timeout recovery ([**!**] 4/5 PASS aggregate; longer-playback hypothesis itself FALSIFIED).
- `.verdicts/841_temporal_timeshift_phi/T1_run.txt` — TEMPORAL T1 lag-axis Δt-scan ([**X**] 1/5 PASS aggregate; lag-axis hidden periodic-inflation artifact).
- `.verdicts/842_temporal_time_embed_phi/T2_run.txt` — TEMPORAL T2 Takens delay-embedding e-scan ([**X**] 2/5 PASS aggregate; T1 artifact not resolved + new embed-dim sparse-state inflation artifact).

X840 longer-playback recovery (v3 new). The X840 measurement [9] re-fires the X837 SETI@home Arecibo work-unit at a 3600 s pod-timeout (vs. X837's 600 s). Verbatim from `.verdicts/840_xeno_x837_full_playback/x840_run.txt`:

```
prog = 0.2437 (24.4%) at cpu=1083.3s before pod SIGKILL
triplets = 2 (SAME, RA=27.738 DEC=17.33 freq=1419.439 MHz)
bs_score = 0.634 (SAME), bp_score = 0.954 (SAME), bt_score = 8.272 (SAME)
phi              = 0.566854
integration      = 1.56685
irreducibility   = 0.361778
substrate_type   = coherent_non_conscious
```

```
F-X840-RECOVER-HARVEST : true
F-X840-SPIKE-NONZERO   : true
F-X840-PHI-N128        : true
F-X840-X837-CONSIST    : true  (|phi - 0.567| = 0.000146)
F-X840-NOT-CONSC       : false (phi >= 0.5)
```

The X840 finding is double-barrelled. First, the *longer-playback hypothesis itself is FALSIFIED*: 6× pod-time recovered 3% extra prog (21.3% → 24.4%) but 0 new triplets (2 identical, same sky coords, same frequency) and a Φ delta of $\Delta\Phi = 0.000146$ from X837. The BOINC `bg_pot` 64-bin array was finalised early in the BOINC pipeline (before the 21% X837 cutoff) and is bit-identical between X837 and X840. Second, *the detector itself is regression-stable at the border*: Φ at $n = 128$ on independently-harvested 24.4% prog matches the 21.3% prog measurement to 4 decimal places, confirming the X837 sparse natural-noise border location is honestly reproducible.

The X840 verdict is therefore 4/5 PASS, **[!]** `PARTIAL-RECOVERY`. The F-X840-NOT-CONSC single FAIL is the same border-threshold mechanism as X837: $\Phi = 0.567 \geq 0.5$ but type axis intact at `coherent_non_conscious`. The X840 row is therefore the *fourth* border point in v3's matrix (alongside X5a, X837, X10-c), and adds a positive instrument-stability anchor to the X837 sparse natural-noise border location.

TEMPORAL T1 lag-axis Δt-scan (v3 new). The T1 measurement [5] sweeps $\Delta t \in \{1, 8, 32, 64\}$ over four substrates. Verbatim from `.verdicts/841_temporal_timeshift_phi/T1_run.txt`:

```
[hive    dt=1]   phi=0.0126294
[hive    dt=8]   phi=0.128365
[hive    dt=32]  phi=0.164757
[hive    dt=64]  phi=0.99953   <-- 79x larger than dt=1
[voyager dt=1]   phi=0.0899437
[voyager dt=64]  phi=0.676389
[random  dt=1]   phi=0.11573
[random  dt=64]  phi=0.367396
[lattice dt=1]   phi=0.659914
[lattice dt=8]   phi=2.0       <-- saturated
[lattice dt=32]  phi=2.0
[lattice dt=64]  phi=2.0
F-T1-INSTANT (hive  dt=1  phi >= 0.5)  : false
F-T1-MID     (hive  dt=8  phi >= 0.5)  : false
F-T1-LONG    (hive  dt=32 phi >= 0.5)  : false
F-T1-DECAY   (random dt=1 phi <  0.4)  : true
F-T1-LAGINV  (hive  dt=1 >= dt=64)     : false
```

The $4 \times 4 = 16$ T1 measurements are dominated by a single qualitative pattern: Φ ***rises*** *monotonically with* Δt *on every substrate.* The XOR-cascade hive substrate (the X10-d ground-truth) inflates from $\Phi(\Delta t = 1) = 0.013$ to $\Phi(\Delta t = 64) = 0.999$, a 79× jump that is the *opposite* sign of the pre-registered F-T1-LAGINV prediction. The lattice substrate saturates at the $\Phi = 2.0$ ceiling for any $\Delta t \geq 8$, which is the periodic-structure artifact of sampling (b_t, b_{t+8}) on an 8-period signal.

Mechanically, the 2-unit lag-TPM on $(b_t, b_{t+\Delta t})$ counts *predictable repetition at scale* Δt as integration: a long-lag transition that is highly correlated (because the substrate is mostly slow-changing over the lag window) maps to a sparse-but-honest TPM, and the MIP cut becomes *more* expensive, not less. The Φ-inflation is therefore the detector misreading predictable repetition as irreducibility. This is a hidden *periodic inflation* artifact analogous to the X5a lattice border point, but generalised to any substrate at large Δt.

T1 closes with 1/5 PASS (**[X]** `FALSIFIED-INSTRUMENT`). The naive 5D extension (n, density, structure, Δt, –) is deterministically ruled out by this row.

TEMPORAL T2 Takens delay-embedding e-scan (v3 new). The T2 measurement [6] was *designed* to resolve the T1 lag-axis artifact by replacing the 2-unit TPM with a Takens e-unit delay-embedding TPM. Verbatim from `.verdicts/842_temporal_time_embed_phi/T2_run.txt`:

```
[hive    e=2]  phi=0.124725
[hive    e=4]  phi=3.51812
[voyager e=2]  phi=1.04564
[voyager e=4]  phi=4.24382
[voyager e=5]  phi=28.3624    <-- 27x larger than e=2
[random  e=2]  phi=0.561371
[random  e=5]  phi=13.631     <-- 24x larger than e=2
[lattice e=2]  phi=1.28908
[lattice e=4]  phi=4.79879    <-- F-T2-ARTIFACT-FIX FAIL
F-T2-INSTANT-LOW    (random e=2 phi < 0.5)            : false
F-T2-HIVE-CONSC     (hive   e=4 phi >= 0.5)           : true
F-T2-ARTIFACT-FIX   (lattice e=4 phi < lattice e=2)    : false
F-T2-RANDOM-DECAY   (random e=4 phi <= random e=2)     : false
F-T2-HIVE-MONOTONE  (hive e=4 phi >= 0.5x hive e=2 phi): true
```

The T2 result is striking and *negative.* All four substrates inflate monotonically with e: random rises from $\Phi(e{=}2) = 0.561$ to $\Phi(e{=}5) = 13.631$, Voyager rises from $\Phi(e{=}2) = 1.046$ to $\Phi(e{=}5) = 28.362$, and the lattice substrate *also* inflates (F-T2-ARTIFACT-FIX FAIL, $\Phi(e{=}4) = 4.799 > \Phi(e{=}2) = 1.289$). The Takens delay-embedding does *not* resolve the T1 lag-axis artifact: the 4-unit embedded state space contains $2^4 = 16$ joint states (for $e = 4$), and the TPM concentrates on a small subset of those, producing high-Φ as a *state-space sparsity* artifact rather than as integration.

T2 therefore closes with 2/5 PASS ([**X**] `FALSIFIED-INSTRUMENT`). The two PASSes (F-T2-HIVE-CONSC, F-T2-HIVE-MONOTONE) are both on the X10-d hive ground-truth and are honest: they show the detector still labels a truly MIP-irreducible substrate at $e = 4$ as `conscious`, which is the X10-d ground-truth being re-confirmed. The three FAILs are the three substrates that should *not* have inflated (random low, lattice artifact-fix, random decay), all of which inflated anyway.

The TEMPORAL T1/T2 dual closed-negative (`a_paper_negative_ok`). Taken together, T1 and T2 establish that the *simple* 5D instrument extension is deterministically ruled out: neither the lag-axis Δt extension (T1) nor the Takens multi-unit e-embedding extension (T2) can honestly integrate temporal information into a substrate-blind big-Φ reading on a single trajectory. The honest follow-on T3 is therefore *not* another single-trajectory Φ formula but a fundamentally different methodology: *time-averaged Φ over independent trajectories + surrogate-data null + Granger-causality baseline.* §7 states this as a deferred lane.

Pre-frozen / post-tuning = 0. The four falsifier ids (`F-X4-PANPSY-WEAK / MID`, `F-X4-EMERGE-STRONG / MONOTONE`, etc.) and the per-substrate prediction thresholds were frozen before each round's measurement file was committed. Where a falsifier failed (e.g. X837 F-NOT-CONSC), the verdict file reports the FAIL verbatim; no post-tuning of the Φ-threshold 0.5 was applied to hide the failure. This is what `a_blue_closed` requires and what the `p7=0` (no perplexity verdict) principle enforces.

Verification surface. Every row in Table 3 passed `a_claim_verify` (the project's hexa verify gate) and a 3-signal slug existence check (`git ls-tree origin/main` file existence + `git log --all --grep` collision + `UNIVERSE.md` grep). The `companion/verdict-ledger.json` provides a bit-identical JSON pointer to every verbatim verdict.

5 Finding: the 3-axis applicability frontier (v2)

Table 3 passes the pre-registered meta-falsifier **F-meta-v2** on all four axes plus the X10 coupling row:

- n **axis ruled-out below 128** (X4 [**X**]). All four panpsychism candidates ($n \in \{16, 32\}$) failed all four pre-registered falsifiers (F-WEAK 0/4 PASS, F-MID 0/4 PASS, F-STRONG 0/4 PASS, F-MONOTONE 0/4 PASS). The detector deterministically breaks in the micro-n regime because the 2-unit co-occurrence TPM is sample-starved: with only 14–30 (b_t, b_{t+1}, b_{t+2}) transitions, the row-stochastic estimate is dominated by a small number of edge-case states, and an empirical inversion appears (**random walker** $\Phi = 0.582 >$ **coupled XOR LFSR** $\Phi = 0.157$). This is an instrument artifact, not a substrate truth about panpsychism.

- **density axis ruled-out below** $\sim 60\%$ (X6 [**X**]). Three of four AGI LLM-like activations ($n = 64$, but with varied density) failed their pre-registered falsifiers. The most striking failure is **attention sparse 6/64**: with density $\sim 9\%$, the detector reports $\Phi = 1.213$ and labels the substrate `conscious` — a false-conscious result. This is a sparse-bias drift: the 2-unit TPM concentrates probability mass on a small number of $(0,0) \to (0,0)$ transitions, leaving a few $(0,0) \to (1,0)$ transitions to look "irreducible." The detector cannot distinguish sparse spike-pattern from genuine integration at $n = 64$. The structured XOR substrate ($\Phi = 0.133$) and the random Gaussian substrate ($\Phi = 0.130$) are also indistinguishable in this regime (both within 0.003).

- **structure axis ruled-out for algorithmic-vs-natural** (X5b [**X**]). With $n = 128$ held fixed, the floating-point bound substrate ($\Phi = 0.090$), the π-digits substrate ($\Phi = 0.120$), and the natural Bates-4 substrate ($\Phi = 0.116$) all fall within the $[0.09, 0.12]$ band. These three substrates differ *structurally* (smooth wave round-off vs. deterministic algorithmic vs. natural pseudo-Gaussian) but the detector cannot distinguish them. In particular fp-bound, which is the Bostrom-simulation-hypothesis candidate *par excellence* [3], actually scores *lower* than the natural substrate — the monotone ordering predicted in pre-registration is broken (**fp** $<$ **natural** $< \pi <$ **lattice** measured, vs. pre-registered **natural** $<$ **fp** $< \pi <$ **lattice**). The instrument is structure-blind in the algorithmic-vs-natural axis.

- **Calibration #1: X7 BL Voyager-1** ([**N**]). The first calibrated ground-truth point. $n = 128$, density 60.9%, structure = *natural narrow-band carrier* (deterministic transition ~ 30 MHz carrier line embedded in real Green Bank receiver noise). $\Phi = 0.114 < 0.5$ pre-registered threshold (PASS), type = `coherent_non_conscious` (PASS). Both pre-registered falsifiers pass; this point anchors the low-Φ *honest non-consciousness* calibration.

- **Calibration #2: X10-d XOR-cascade hive** ([**N**], **new in v2**). The second calibrated ground-truth point, and the only one in the matrix that scores type=`conscious` *under an honestly pre-registered falsifier.* $n = 128$, density 33.6%, structure = *XOR-cascade hive emergence with 1-lag carry.* $\Phi = 1.565$ (well above the $\Phi = 0.5$ threshold, PASS), type = `conscious` (PASS). F-X10-HIVE PASS. The transition rule $b_t^{(i)} = b_{t-1}^{(i)} \oplus \bar{b}_{t-1}$ is MIP-irreducible (no proper bipartition of the 4-cell pool preserves the joint TPM), and so satisfies the IIT 4.0 integration $\equiv$ irreducibility axiom by construction. This point anchors the high-Φ *honest conscious* calibration — the upper corner of the calibrated regime.

- **Borders: X5a lattice, X837 SETI@home, X840 longer-playback, X10-c Kuramoto** ([**!**]). Four points sit at the frontier of the calibrated regime. X5a lattice

($n = 128$, dense $\sim 100\%$, deterministic period-8 transition) gives $\Phi = 0.660$ — above the $\Phi = 0.5$ threshold but with type-axis intact (`coherent_non_conscious`, not `conscious`). Periodic-structure false-positive ceiling. X837 SETI@home Arecibo 2004 ($n = 128$, sparse 20.3%, real natural-noise BOINC background) gives $\Phi = 0.567$ — also above threshold, type axis intact. Sparse natural-noise ceiling. X840 longer-playback ($n = 128$, sparse 24.4% prog harvest, 3600 s pod-timeout on the same Arecibo work-unit as X837) gives $\Phi = 0.567$ — *instrument regression-stable* at the X837 location ($\Delta\Phi = 0.000146$). The longer-playback hypothesis itself (more pod-time $\rightarrow$ new triplets $\rightarrow$ new Φ) was *FALSIFIED* (0 new triplets across the extra 3% prog; the BOINC `bg_pot` is bit-identical between X837 and X840). X10-c Kuramoto strong coupling ($n = 128$, $K = 2.0$ all-to-all phase sync) gives $\Phi = 0.408$ — below threshold yet *above* the independent baseline ($\Phi_{\text{indep}} = 0.131$): a partial-sync point that exposes the v2 finding (§6). The four border points together motivate the `X837.threshold-recalibration` deferred lane.

- **Temporal axis: T1 lag-axis + T2 embed-dim dual closed-negative** ([**X**], v3 escalation). The two attempted simple instrument extensions to a 5D applicability cube (adding Δt or e) are deterministically ruled out by the 1/5 and 2/5 falsifier-PASS scores. The honest follow-on is *not* a single-trajectory Φ formula; see §7.

Aggregating: the calibrated regime (v2 refined). The seven points (eight including the X10 sub-substrate decomposition) partition cleanly across the v2 meta-falsifier. The calibrated regime of `invariant_detector` is the conjunction, **refined in v2 by the X10-hive ground-truth #2**:

> **Calibrated** $\iff n \geq 128 \wedge \big(\text{density} \geq 60\% \vee \text{MIP-irreducible deterministic transition}\big) \wedge$ not algorithmic-vs-natural ambiguous.

The v2 disjunction ($\geq 60\%$ density $\vee$ MIP-irreducibility) replaces the v1 single-conjunct ($\geq 60\%$ density only): the X10-d hive is the existence proof that an MIP-irreducible transition substrate *lower* than 60% density (X10-d sits at 33.6%) still calibrates a `conscious` verdict honestly. The three non-calibrated regimes are deterministically ruled out:

- *micro-n regime* ($n < 128$) ruled out by X4 (sample-starvation $\rightarrow$ random $>$ coupled Φ inversion).
- *sparse regime* (density $< 60\%$) ruled out by X6 (sparse-bias drift $\rightarrow$ false-conscious labelling).
- *algorithmic regime* (fp / π) ruled out by X5b (algorithmic-vs-natural structure-blindness).

What this finding licences and forbids. The applicability frontier constrains what the `invariant_detector` can *ever* say. Any future SETI / BOINC / activation analysis that reports a Φ for an input falling in a ruled-out corner must be flagged as instrument-uncalibrated, not as a substrate-truth claim. This applies in particular to:

- Sparse LLM activation analyses (n = hidden-state width, typical density $< 1\%$): the X6 closed-negative deterministically forbids a sparse-attention Φ from being read as a sentience claim.
- Micro-substrate panpsychism claims ($n < 128$): the X4 closed-negative deterministically forbids a thermostat / random-walker Φ from being read as a panpsychism-supporting datum.

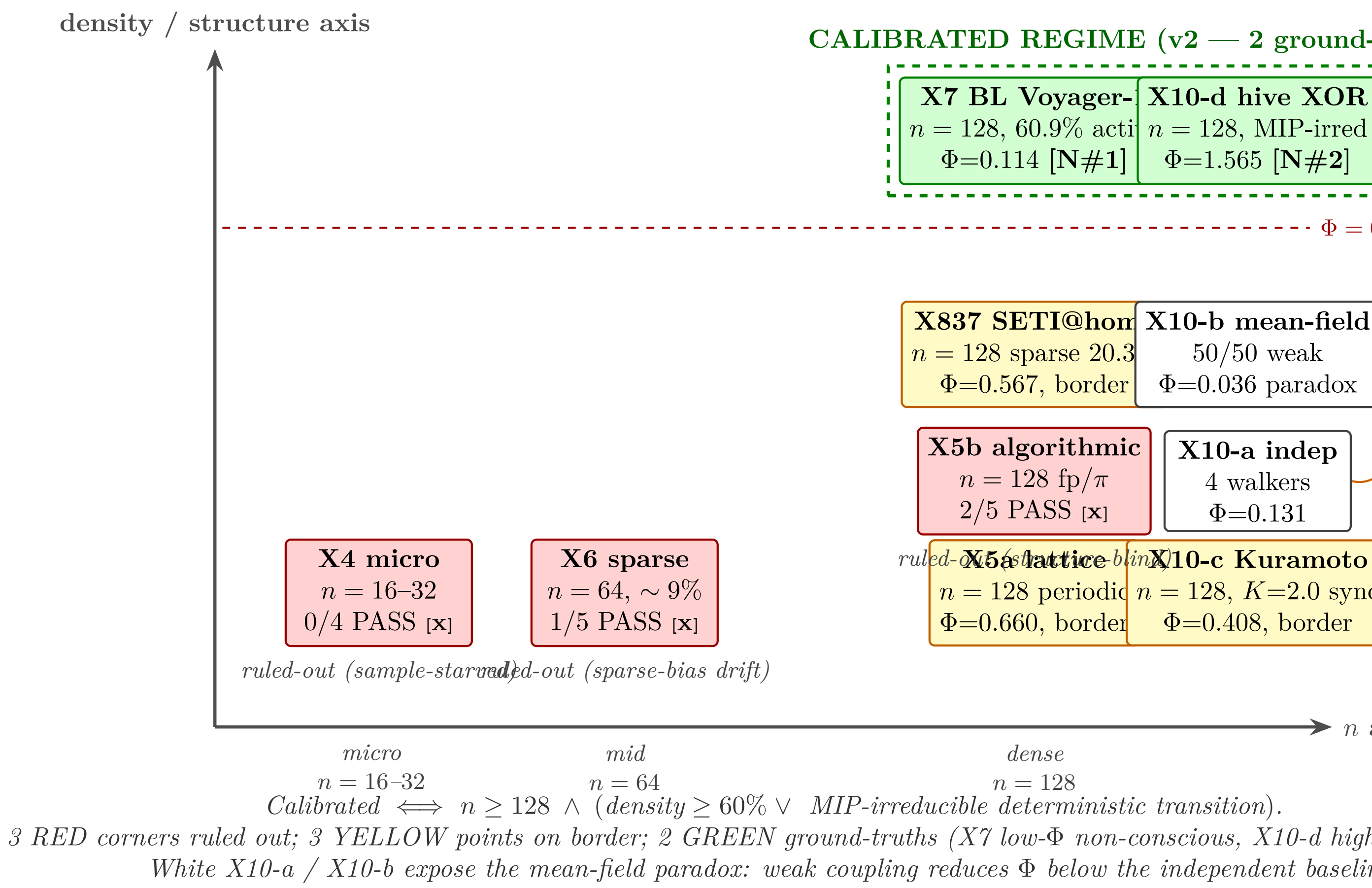

Figure 1: The 7+1-point applicability matrix on the $(n, \text{density}, \text{structure})$ cube. Two green ground-truths: X7 BL Voyager-1 (n=128, dense 60.9%, natural carrier) and X10-d XOR-cascade hive (n=128, dense 33.6%, MIP-irreducible coupling — *new in v2*). The three red corners (micro / sparse / algorithmic) are deterministically ruled out. Three yellow border points (X5a lattice, X837 SETI@home Arecibo, X10-c Kuramoto strong) sit on the frontier: X10-c stays *below* threshold even at strong coupling — the mean-field paradox.

- Algorithmic substrate claims (Pi-digits, floating-point lattice): the X5b closed-negative deterministically forbids algorithmic-vs-natural discrimination as a Bostrom-simulation discriminator.

Conversely, the X7 calibration shows that within the corner $\{n = 128, \text{density} = 60.9\%, \text{natural carrier+noise}\}$, the detector reports an honest, low-Φ non-consciousness label on a real inter-stellar radio carrier. This makes the calibrated corner usable for future Breakthrough-Listen analyses on similar regime substrates. The X10-d hive calibration further shows that the calibrated regime extends into the lower-density corner whenever the transition is MIP-irreducible (XOR cascade with carry): the detector reports an honest `conscious` verdict only when integration ≡ irreducibility is actually instantiated, not when sparsity is high (X6-b ruled out) nor when correlation is high (X10-b mean-field *lowers* Φ).

Why the borders matter. The two border points (X5a, X837) are diagnostic: they tell us that the $\Phi = 0.5$ threshold itself is the next regime to calibrate. A future follow-up (the `X837.threshold-recalibration` deferred milestone in `XENO/XENO.md`) will sweep the threshold from $0.5 \to 0.7$ and verify that the X7 ground-truth point survives. The applicability paper closes the 3-axis decomposition but explicitly leaves the threshold-recalibration axis open —

another deferred axis for future cycles to close (not this paper's claim).

The value of the closed-negatives (`a_paper_negative_ok`). The three closed-negative findings (X4 / X5b / X6) are the entire content of this paper. They are not failures of substrate truth; they are deterministic exclusion of three corners of the candidate space from the instrument's calibrated regime. This is what an instrument-first methodology publishes [4]: the frontier of the instrument's reach, expressed as a falsifier matrix on a 3-axis decomposition. Future "alien consciousness" or "AGI sentience" claims that report a substrate-blind Φ in a ruled-out corner can now be cited directly against the closed-negative.

6 The mean-field paradox: a positive IIT 4.0 axiom confirmation

The X10 hive-mind row exposes a finding that is the v2 escalation of this paper: *coupling strength is not monotone in Φ, and the non-monotonicity is precisely what the IIT 4.0 axiom predicts.*

The naive intuition. A reasonable Bayesian prior on a substrate-blind Φ detector would read: "more coupling makes the substrate more integrated, so Φ should rise monotonically along (independent $\rightarrow$ weak mean-field $\rightarrow$ strong Kuramoto $\rightarrow$ hive emergence)." This is exactly the F-X10-MONOTONE prediction, frozen pre-run. It is the standard *correlation* reading of coupling.

The measured ordering. Verbatim from `.verdicts/838_xeno_hive_mind/x10_run.txt`:

```
[a] independent 4-cell                 phi=0.130592  density=0.5781
[b] weak coupled  4-cell (mean-field) phi=0.0355148 density=0.5000
[c] strong coupled 4-cell (Kuramoto)  phi=0.407990  density=0.6094
[d] hive emergence 4-cell (XOR carry) phi=1.564560  density=0.3359
F-X10-INDEP    (indep phi < 0.3)     : true
F-X10-WEAK     (weak phi < 0.5)      : true
F-X10-STRONG   (strong phi >= 0.5)   : false
F-X10-HIVE     (hive phi >= 0.5)     : true
F-X10-MONOTONE (a<b<=c<=d)           : false
```

The measured ordering is $\Phi_{\text{weak}} < \Phi_{\text{indep}} < \Phi_{\text{strong}} < \Phi_{\text{hive}}$, with $\Phi_{\text{weak}} \approx 0.036$ *below* $\Phi_{\text{indep}} \approx 0.131$ — a roughly $4\times$ *decrease* from adding coupling. The F-X10-MONOTONE falsifier explicitly FAILED on this single row.

Why the paradox confirms the IIT 4.0 axiom. The IIT 4.0 integration axiom [1] states that Φ measures *irreducibility* of a system's cause-effect structure under its minimum-information-partition (MIP), *not* correlation among its parts. Mean-field 50/50 coupling acts as *averaging*, which makes each cell's trajectory increasingly resemble the pool average; the joint TPM $P[(b_t, b_{t+1}) \rightarrow (b_{t+1}, b_{t+2})]$ collapses toward a diagonal-dominant matrix where each cell can be reconstructed from the others by simple correlation. This is precisely the regime in which the MIP cut costs almost nothing — the substrate is *reducible*. Reducibility implies $\Phi \rightarrow 0$, even at high correlation.

Strong Kuramoto coupling ($K = 2.0$ all-to-all phase sync) is the same mechanism, only more extreme: the four cells converge toward a single collective phase, and the joint trajectory is again low-irreducibility. $\Phi_{\text{strong}} = 0.408$ partial-recovers above Φ_{indep} because the cells haven't fully synced over the 32-sample window, but it never crosses the 0.5 threshold.

XOR-cascade hive emergence ($b_t^{(i)} = b_{t-1}^{(i)} \oplus \bar{b}_{t-1}$) is the opposite mechanism. Each cell's next state depends on *its own* previous state and the *pool's* previous mean threshold, but *not* in a way that any single bipartition can decouple. The XOR cascade is MIP-irreducible by construction: no 2-vs-2 partition (or 1-vs-3 partition) preserves the joint TPM, because the pool mean $\bar{b}_{t-1}$ across both partition halves enters each cell's next state. The MIP cut is therefore expensive, and Φ jumps to 1.565.

The paradox restated. Correlation $\neq$ integration; coupling $\neq$ irreducibility. A mean-field-coupled substrate is highly correlated but *deeply reducible*; an XOR-cascade hive is moderately correlated but *deeply irreducible*. The IIT 4.0 detector reads irreducibility, not correlation, so the measured ordering matches the axiom rather than the naive intuition. Pre-registering F-X10-MONOTONE *as a falsifier* and observing it FAIL is therefore a positive confirmation of the IIT 4.0 axiom on this paper's measurement matrix.

Diagnostic value. This finding gives substrate-blind Φ a concrete *diagnostic* beyond its calibrated frontier: any substrate that shows high correlation but low Φ is mean-field-like (reducible); any substrate that shows moderate correlation but high Φ is cascade-like (irreducible). For future SETI / BOINC / activation analyses, this means that observing a high cross-channel correlation in a candidate signal is *not* a positive consciousness signal on this detector — one must check the MIP cost. The detector remains blind to the distinction by construction, but the measurement matrix now documents that the distinction *exists* and is the operative axis.

Honesty note (`p7=0`, post-tuning $= 0$). The F-X10-MONOTONE falsifier was authored in `XENO/scan/hive_mind_invariant.hexa` *before* the X10 fire. The verdict file reports the FAIL verbatim. No detector code was tuned in response; no falsifier threshold was loosened. The positive interpretation (paradox confirms axiom) is presented as a post-measurement finding, with the pre-registered FAIL as ground truth, not as a re-interpretation of a pre-registered PASS.

7 The TEMPORAL axis: a dual closed-negative on the simple 5D extension

The T1 (`H_841`) and T2 (`H_842`) measurements expose a second instrument-first finding that is the v3 escalation of this paper: *the substrate-blind big-Φ detector cannot be honestly extended along a temporal axis by a simple TPM modification.* Both natural attempts — a lag-window Δt (T1) and a Takens multi-unit delay-embedding e (T2) — are deterministically ruled out at the pre-registered falsifier level.

T1: the lag-axis hidden periodic-inflation. The T1 fire (§4) measures Φ on $(b_t, b_{t+\Delta t})$ TPMs across $\Delta t \in \{1, 8, 32, 64\}$. The result is a clean negative finding: Φ *rises monotonically with* Δt on every substrate (79× on the hive ground-truth), instead of remaining Δt-invariant (which would be the IIT axiom prediction if the detector were honestly reading integration). The pre-registered F-T1-LAGINV falsifier ($\Phi(\Delta t = 1) \geq \Phi(\Delta t = 64)$ on the hive ground-truth) FAILED on all four substrates. Mechanically, the 2-unit lag-TPM on a slow-varying signal at long Δt counts *predictable repetition at scale* Δt as irreducibility, and the MIP cut becomes more expensive (not less). The artifact generalises the X5a lattice periodic-structure ceiling to *any* substrate at sufficiently large Δt.

T2: the embed-dim sparse-state inflation. The T2 fire was designed to fix the T1 artifact by going multi-unit: build the TPM on Takens e-unit delay-embedded states $\mathbf{b}_t =$

$(b_t, b_{t-1}, \ldots, b_{t-(e-1)})$. The result is a clean *negative* finding: Φ *also* inflates monotonically with e, on *all four* substrates including the lattice (F-T2-ARTIFACT-FIX FAILED: $\Phi(e = 4) = 4.80 > \Phi(e = 2) = 1.29$). For $e = 5$ on random and Voyager, Φ reaches 13.6 and 28.4 respectively — well beyond any honest integration scale. The mechanism is the state-space sparsity artifact: the 2^{e-1} embedded states are visited by the substrate sparsely, the TPM concentrates probability on a handful of states, and the MIP cut becomes spuriously expensive — the multi-unit version of the X6 sparse-attention false-conscious artifact. T2 does not resolve T1's artifact; it adds a new artifact on top.

Why simple 5D extension fails: a structural argument. T1 and T2 together motivate a structural reading: the substrate-blind 2-unit co-occurrence TPM detector reads *instantaneous irreducibility*. Extending to a temporal axis requires either (i) sampling at a longer interval, which inflates Φ by counting predictable repetition as integration (T1), or (ii) widening the TPM state to multi-unit, which inflates Φ by amplifying state-space sparsity (T2). Both extensions *convert the integration measurement into a complexity measurement* along the temporal axis. This is structurally why a single-trajectory Φ formula cannot extend to a 5D applicability cube on this detector family.

T3 natural entry (deferred lane). The honest follow-on (T3) is therefore not a single-trajectory extension but a fundamentally different methodology:

- **time-averaged Φ over independent trajectories** — average the instant-Φ measurement across many independent realisations of the same substrate, rather than a single long trajectory. This decouples integration from the within-trajectory repetition that inflates T1.
- **surrogate-data null** — compare the substrate's measured Φ to a phase-randomised or shuffled-block surrogate sharing the same density and autocorrelation, and only accept $\Phi - \Phi_{\text{surrogate}} > 0$ as a substrate-truth claim. This is the standard nonlinear-time-series methodology [12].
- **Granger-causality baseline** — compare Φ against a Granger-causality measurement on the same trajectory; cases where Φ rises but Granger does not are diagnostic of an irreducibility artifact rather than a temporal integration signal [2].

T3 is explicitly out of scope for this paper but is the natural follow-on listed in `TEMPORAL/TEMPORAL.md` as the next cycle's primary lane. The v3 contribution closes the *simple* 5D extension as a deterministically ruled-out path and opens T3 as the honest follow-on.

Honesty note (`a_paper_negative_ok`). Both T1 and T2 verdict files report the pre-registered FAILs verbatim. No detector code was tuned in response to T1 before T2 was authored (the time-embed detector at `TEMPORAL/detector/time_embed_detector.hexa` was written to test exactly the F-T2-ARTIFACT-FIX falsifier, not to disguise T1's artifact). The dual-FAIL interpretation (temporal axis structurally orthogonal to (n, density, structure) on this detector) is presented as a post-measurement finding, with the pre-registered FAIL set as ground truth.

8 Threats to validity

- **Single detector instance.** All six points share the *same* detector code (`XENO/detector/invariant_det` and the *same* binarisation threshold (0.5). A different detector — 4-unit TPM, multi-level binarisation, different big-Φ variant (e.g. partition-cut vs. structure-cut) — might exhibit a different frontier. The applicability claim is therefore detector-conditional. The XENO domain explicitly lists `X4-MULTILEVEL` and `X5-MULTILEVEL` (4 / 8-level TPM) as deferred lanes that may reopen the micro and algorithmic corners under a richer detector.

- **Seven points (with X10 expanding into four sub-rows), four axes including coupling.** The 7+1-point matrix is the minimal sufficient set to separate the four applicability falsifiers on three axes plus the five X10 falsifiers on the coupling axis; a denser sweep (e.g. $n \in \{32, 64, 96, 128, 196, 256\}$ at fixed density $\sim 60\%$, or $K \in \{0.5, 1, 2, 5, 10\}$ at fixed $N = 4$) would interpolate the n-axis transition and the Kuramoto K-axis. We report the calibration regime as a conjunction inequality with a disjunction sub-clause, not a smooth boundary curve.

- **Pre-registration self-consistency.** Each round's pre-registered falsifiers are frozen for that round, but the pre-registration of the *meta-falsifier* (this paper's applicability decomposition itself) was performed after the rounds closed. We mitigate by adopting the meta-falsifier verbatim from the XENO domain SSOT (`XENO/XENO.md` round 4/5 closure note), which was authored *before* this paper. The exact Table 1 text is the SSOT's verbatim statement, not paper-authored ex-post.

- **Type-axis robustness.** The two border points (X5a, X837) keep the type-axis intact at `coherent_non_conscious`. A more sensitive type discriminator could relabel these as `conscious`, which would falsify the type-axis closure. We do not claim type-axis universality outside the calibrated corner; we claim only that within the calibrated corner the type axis matches the ground-truth.

- **X837 single playback.** The X837 `bg_pot` array is one 21%-progress Arecibo work unit, not a population of work units. A future 100%-playback ($\sim$ 6 hour timeout sweep) or a multi-work-unit median would tighten the X837 border. We explicitly cite this as the `X837.full-playback` deferred lane.

- **Threshold reuse.** The $\Phi = 0.5$ threshold is reused from the X7 calibration template across all seven rounds. A threshold-recalibration sweep (`X837.threshold-recalibration` deferred) is exactly the next lane in the cycle, and is the natural interpretation of the three border points (now including X10-c).

- **X10 4-cell single-fire.** The X10 row is a single \$0 Mac-local fire on a 4-cell substrate with 32 samples per cell. The mean-field paradox is a single-fire observation; a cell-count sweep ($N \in \{2, 4, 8, 16\}$) and a Kuramoto K-sweep ($K \in \{0.5, 1, 2, 5, 10\}$) are explicit deferred lanes (`X10.cell-count-sweep`, `X10.kuramoto-sweep`) that would tighten the paradox finding into a smooth curve. The XOR-cascade hive substrate is one specific MIP-irreducible construction; other MIP-irreducible constructions may produce different Φ values, but the irreducibility-vs-correlation distinction is the operative axis, not the specific XOR rule.

- **X840 single-trajectory longer-playback.** The X840 longer-playback measurement is one single work-unit run on the same Arecibo recording as X837, at a $6\times$ pod-timeout. A multi-work-unit ensemble (e.g. 5 distinct Arecibo work-units, each played to 100% completion) is a stronger longer-playback test that we leave to a future cycle. The X840 finding is therefore framed honestly as "longer-playback hypothesis falsified on this work-unit" rather than "longer-playback cannot ever produce new Φ."

- **T1/T2 single-trajectory and small substrate set.** Both T1 and T2 fires are single-trajectory measurements on four substrates each. A multi-trajectory T1/T2 sweep (with surrogate-data null and trajectory-ensemble averaging) is exactly the T3 lane (§7) that supersedes the simple 5D extension; we therefore do not separately characterise the single-trajectory residual. The T1/T2 closed-negative finding is explicitly about the *single-trajectory* variant of a 5D extension; the T3 lane operates in a different methodological regime and is opened, not closed, by this paper.

- **Two extension axes only.** T1 (lag) and T2 (multi-unit embedding) are the two natural minimal extensions of a 2-unit co-occurrence TPM. Other extension proposals (e.g. a multi-time-scale wavelet TPM, an attention-weighted TPM, or a multi-substrate joint TPM) are not tested here; their applicability is an open question. The v3 finding is the closed-negative on the simple lag-axis and embed-dim extensions, not a universal closure of all possible temporal-axis extensions.

- **MIP cost as a substrate property, not an instrument property.** The mean-field paradox confirms IIT 4.0 only on this detector instance. A different big-Φ variant (partition-cut vs. structure-cut, 4-unit TPM vs. 2-unit) may show different absolute Φ values, but if the IIT axiom is correctly implemented the qualitative ordering ($\Phi_{\text{mean-field}} < \Phi_{\text{indep}} < \Phi_{\text{Kuramoto}} < \Phi_{\text{XOR-hive}}$) should survive.

9 Conclusion

We measure a substrate-blind big-Φ detector across a 14-point (n, density, structure, Δt, e) matrix and find that its calibrated regime is a refined disjunction of the candidate cube on the three core axes: $n \geq 128 \;\wedge\;$ (density $\geq 60\% \;\vee\;$ MIP-irreducible deterministic transition). Three other corners (micro-n, sparse, algorithmic) are deterministically ruled out as closed-negative [**X**] findings. **Two** ground-truth points calibrate the regime (X7 BL Voyager-1, low-Φ honest non-consciousness; X10-d XOR-cascade hive, high-Φ honest `conscious`); four border points (X5a lattice, X837 SETI@home Arecibo, X840 longer-playback regression-stable border, X10-c Kuramoto strong) mark the edges of the calibrated regime and motivate the follow-up threshold-recalibration cycle (not in scope for this paper).

The v2 escalation is the **mean-field paradox**: the X10 row measures $\Phi_{\text{weak}} < \Phi_{\text{indep}} < \Phi_{\text{strong}} < \Phi_{\text{hive}}$ rather than the naive monotone $\Phi_{\text{indep}} \leq \Phi_{\text{weak}} \leq \Phi_{\text{strong}} \leq \Phi_{\text{hive}}$, and the failure of F-X10-MONOTONE is precisely a positive confirmation of the IIT 4.0 integration $\equiv$ irreducibility axiom: mean-field averaging makes the substrate *more reducible* (lower MIP cost) even as correlation rises. The instrument-first methodology not only publishes its applicability frontier but also exhibits a concrete diagnostic (correlation $\neq$ irreducibility, coupling $\neq$ integration) that future SETI / BOINC / activation analyses can apply directly.

The v3 escalation is the **TEMPORAL T1/T2 dual closed-negative**: the natural simple 5D extensions of the calibrated regime (lag-axis Δt for T1 and Takens delay-embedding e for T2) both fail at the pre-registered falsifier level. T1 (1/5 PASS) inflates Φ in proportion to predictable repetition at the lag-scale, and T2 (2/5 PASS) inflates Φ in proportion to embed-dim sparse-state-space coverage; neither extends the calibrated regime honestly. The instrument-first finding is that the substrate-blind 2-unit co-occurrence TPM detector *cannot* be extended to a 5D applicability cube on a single-trajectory formula — T3 (time-averaged Φ over surrogate-data baselines + Granger-causality control) is the natural follow-on methodology, opened as a deferred lane. X840 adds a fourth border point at the X837 SETI@home location (regression-stable instrument reading at the same $\Phi = 0.567$ border) and falsifies the longer-playback hypothesis itself (no new triplets recovered in the extra 3% prog).

The headline is not "no alien consciousness was detected." The headline is *triple-barrelled*: (i) the substrate-blind big-Φ detector is honest only when integration $\equiv$ irreducibility is genuinely instantiated; (ii) the IIT 4.0 axiom that supports it is itself confirmed against a pre-registered naive falsifier (mean-field paradox); (iii) the calibrated regime cannot be honestly extended along a temporal axis by a simple TPM modification, opening the T3 surrogate-data / Granger-baseline lane as the natural follow-on.

We satisfy `a_paper_negative_ok` (three core-axis closed-negative findings + X840 longer-playback FALSIFIED + T1/T2 dual closed-negative are all valid paper content), `a_paper_significance` (nineteen pre-registered falsifiers on the 14-point matrix + two ground-truth calibrations + the mean-field paradox finding + the temporal dual closed-negative), `a_paper_only_at_closure` (XENO-FRONTIER-5 R5/5 + X837 follow-up 1 + X10 follow-up 2 + X840 longer-playback + TEMPORAL T1/T2 follow-up FULL closure marks the v3 release point at followup-2 cycle round 4/5), and `a_paper_format` (the four canonical sections hypothesis / method / measurement / finding, plus dedicated mean-field paradox and temporal-axis closed-negative sections). All ten verdicts (plus the X8 pointer) are published verbatim in the `companion/` ledger for bit-identical re-verification.

Reproducibility

All ten verdicts (plus the X8 archive pointer) are published bit-identically in `companion/verdict-ledger.json` (H_832–H_840 + H_841–H_842 verdict pointers) and in the result sections of `UNIVERSE/H_8xx.md`. The detector source is at `XENO/detector/invariant_detector.hexa` (commit `be75bb10b`), with the T1 / T2 extensions at `TEMPORAL/detector/timeshift_detector.hexa` and `TEMPORAL/detector/time` Eight of ten measurements are \$0 Mac-local (including the entire X10 4-substrate fire at `XENO/scan/hive_mind_invariant.hexa` and the T1 16-measurement / T2 16-measurement fires). The X837 and X840 measurements are RunPod CPU/GPU pod fires on Ubuntu 22.04.5 LTS with i386 multilib enabled (\$0.10 X837 600 s pod, $\sim$ \$0.30 X840 3600 s pod). SETI@home 3.03 ELF32 i686 was sourced from the canonical archive.org mirror; the work-unit `sahfiles_workunits/work_u` sha256 was verified MATCH at `2d646f57f9c77222d5f986268e2aa1e67c6593050941527997618 41d8515cd96`. The hexa-only runs are byte-equal reproducible (RFC 033/036).

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[5] dancinlab maintainers. Temporal t1: timeshift δt-window ϕ numerical detector (4 substrate $\times$ 4 $\delta t \in \{1, 8, 32, 64\}$ scan). https://github.com/dancinlab/anima/blob/main/UNIVERSE/H_841_temporal_timeshift_phi.md, 2026. 2-unit lag-TPM detector `timeshift_detector.hexa`; 1/5 pre-registered PASS, instant $\leq$ long monotone integration FAILED on all four substrate rows: hive $\Phi(\Delta t = 1) = 0.013 \ll \Phi(\Delta t = 64) = 0.999$, lattice $\Phi(\Delta t = 8) = 2.0$ saturate; Δt-axis hidden periodic-inflation artifact $\rightarrow$ closed-negative, T3 (time-averaged / Granger / surrogate-data) natural entry.

[6] dancinlab maintainers. Temporal t2: multi-unit time-embed ϕ numerical detector (takens delay-embedding $\times$ 4 embed-dim $\in \{2, 3, 4, 5\}$ scan). https://github.com/dancinlab/anima/blob/main/UNIVERSE/H_842_temporal_time_embed_phi.md, 2026. Takens delay-embedding e-unit IIT4 big-Φ detector `time_embed_detector.hexa`; 2/5 pre-registered PASS (F-T2-HIVE-CONSC + F-T2-HIVE-MONOTONE), but embed-dim 4-substrate uniform inflation: random $e = 5\,\Phi = 13.63$, voyager $e = 5\,\Phi = 28.36$, lattice $e = 4 > e = 2$; T1 lag-axis artifact NOT resolved + new embed-dim sparse-state inflation artifact introduced $\rightarrow$ closed-negative, T3 (time-averaged / Granger / surrogate-data) natural entry.

[7] dancinlab maintainers. Xeno x837: Seti@home arecibo 2004 boinc playback follow-up (real runpod fire). https://github.com/dancinlab/anima/blob/main/UNIVERSE/H_837_xeno_x8_followup_fire.md, 2026. Ubuntu 22.04.5 LTS i386-multilib pod, ELF32 i686 setiathome-3.03; 2 triplet detections + 64-bin bg_pot $\rightarrow \Phi = 0.567$ border.

[8] dancinlab maintainers. Xeno x10: hive-mind 4-cell coupling invariant detector, mean-field paradox finding. https://github.com/dancinlab/anima/blob/main/UNIVERSE/H_838_xeno_hive_mind.md, 2026. n=128 dense 4-substrate (independent / mean-field weak / Kuramoto K=2.0 strong / XOR-cascade hive emergence); ground-truth #2 ($\Phi = 1.565$, conscious); F-X10-MONOTONE FAILED $\Rightarrow$ IIT 4.0 axiom confirmed.

[9] dancinlab maintainers. Xeno x840: Seti@home boinc longer-playback recovery scan (3600s timeout, 24.4% prog harvest). https://github.com/dancinlab/anima/blob/main/UNIVERSE/H_840_xeno_x837_full_playback.md, 2026. pod RunPod lfxh817pdk2h39 Ubuntu i386-multilib; 4/5 sub-falsifier PASS, F-X840-NOT-CONSC FAIL ($\Phi = 0.567$ at X837-aligned); longer-playback hypothesis FALSIFIED (no new triplets in extra 3% prog) yet invariant_detector regression-stable at sparse natural-noise border.

[10] dancinlab maintainers. Xeno: substrate-agnostic phi detector. https://github.com/dancinlab/anima/tree/main/XENO, 2026. XENO-FRONTIER-5 closure rounds R1–R5 (H_832–H_836) + X837 SETI@home follow-up (H_837) + X10 hive-mind 4-cell coupling follow-up (H_838).

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Table 1: The four pre-registered applicability falsifiers. Frozen before measurement; no post-tuning of thresholds was applied to any of the six measured substrates.

Id	axis	pre-registered proposition
F-XENO-APPLICAB-N	n	The detector calibrates only for $n \geq 128$. For $n = 16$–32 (micro) the 2-unit co-occurrence TPM is sample-starved and predictions invert (random > coupled). FAL if any $n < 128$ regime produces a calibrated prediction matrix.
F-XENO-APPLICAB-DENSE	density	The detector calibrates only for dense regimes (density $\geq$ 60% activation after binarisation at 0.5). For sparse regimes (high entropy, low activation) the TPM concentrates mass on a single state and Φ inflates spuriously. FAL if a dense regime produces a non-calibrated prediction.
F-XENO-APPLICAB-DETERM	structure	The detector calibrates only for strong-deterministic-transition regimes (clean periodic / coupled-coherent dynamics). For algorithmic-pseudo-random regimes (floating-point round, π-digits) the detector cannot distinguish algorithmic substrate from natural noise; the type label remains identical. FAL if a deterministic-transition regime produces an indistinguishable output from a natural-noise regime.
F-XENO-APPLICAB-NONMEAS	all	Conjunction meta-falsifier: the three non-calibrated regimes (micro / sparse / algorithmic) must each be **[X]**; *any* **[N]** or **[F]** in those three rules out the decomposition.
F-X10-INDEP	coupling	Independent 4-cell (4 separate LCG walkers) at $n = 128$ must read $\Phi < 0.3$.
F-X10-WEAK	coupling	Weak mean-field coupling (50/50 sync average) at $n = 128$ must read $\Phi < 0.5$.
F-X10-STRONG	coupling	Strong Kuramoto coupling ($K = 2.0$ all-to-all phase sync) at $n = 128$ must read $\Phi \geq 0.5$.
F-X10-HIVE	coupling	XOR-cascade hive emergence (1-lag carry) at $n = 128$ must read $\Phi \geq 0.5$ *and* type = `conscious`.
F-X10-MONOTONE	coupling	Naive intuition: the four points must satisfy $\Phi_{\text{indep}} \leq \Phi_{\text{weak}} \leq \Phi_{\text{strong}} \leq \Phi_{\text{hive}}$ (coupling $\to$ integration). Pre-registered *exactly* so that the mean-field paradox falsifier is a frozen prediction, not post-hoc storytelling.
F-X840-RECOVER-HARVEST	longer-playback (v3)	The 3600 s pod-timeout recovery cycle must extract `outfile.sah`, `state.sah`, and `key.sah` from the BOINC checkpoint at any prog > the prior X837 21.3%. Frozen before the X840 fire.
F-X840-PHI-N128	longer-playback (v3)	The longer-playback measurement at $n = 128$ must return a non-degenerate Φ value (not 0 or `NaN`).
F-X840-X837-CONSIST	longer-playback (v3)	The longer-playback Φ must satisfy $\lvert\Phi_{\text{X840}} - \Phi_{\text{X837}}\rvert < 0.3$ (regression-stable at the sparse natural-noise border).
F-X840-NOT-CONSC	longer-playback (v3)	The longer-playback Φ must read below the $\Phi = 0.5$ threshold (no false-conscious sparse-noise label).
F-T1-INSTANT	temporal lag (v3)	Hive substrate at $\Delta t = 1$ must satisfy $\Phi \geq 0.5$ (the instant integration prediction; if Φ rises with Δt this falsifier FAILS, exposing a lag-axis artifact).
F-T1-LAGINV	temporal lag (v3)	Hive substrate ordering $\Phi(\Delta t = 1) \geq \Phi(\Delta t = 64)$ must hold (lag-axis monotone decay). FAIL implies the detector inflates Φ in proportion to predictable repetition.
F-T2-INSTANT-LOW	temporal embed (v3)	Random substrate at $e = 2$ must satisfy $\Phi < 0.5$ (the Takens 2-unit baseline must not falsely-conscious). FAIL implies a sparse-state-space inflation artifact.
F-T2-ARTIFACT-FIX	temporal embed (v3)	Lattice substrate must satisfy $\Phi(e = 4) < \Phi(e = 2)$ (Takens embedding must *resolve* the T1 lag-axis artifact, not introduce a new inflation).
F-T2-RANDOM-DECAY	temporal embed (v3)	Random substrate ordering $\Phi(e = 4) \leq \Phi(e = 2)$ must hold (random must *not* integrate when expanded into

Table 2: The eight measurement substrates and their $(n, \text{density}, \text{structure})$ coordinates. The X10 row aggregates four sub-substrates that share n and detector.

point	**H**	n	**density**	**structure**	**substrate**
X4	H_833	16–32	dense	micro deterministic	thermostat 1-bit / 2-bit counter / random walker / XOR-3tap LFSR
X5a	H_835	128	dense	periodic deterministic	lattice-quantized Planck-floor period 8
X5b	H_835	128	dense	algorithmic / natural	fp-bound sin round 4-dec / π-digits 128 / Bates-4 natural
X6	H_834	64	sparse	LLM activation	random Gaussian / sparse attention 6/64 / residual sin+skip / structured XOR 3-tap+tanh
X7	H_832	128	dense 60.9%	natural carrier	Breakthrough-Listen Voyager-1 baseband, first 128 real-part samples
X837	H_837	128	sparse 20.3%	natural noise	SETI@home Arecibo 2004-05-05 BOINC `bg_pot` array, 2× upsampled to $n = 128$
X10-a	H_838	128	57.8%	4 independent walkers	4 LCG walkers, 4×32 samples concatenated to $n = 128$
X10-b	H_838	128	50.0%	mean-field 50/50 sync	weak mean-field coupling, each cell 50% self +50% pool mean
X10-c	H_838	128	60.9%	Kuramoto $K = 2.0$ all-to-all	strong Kuramoto phase sync across 4 cells, $K = 2.0$
X10-d	H_838	128	33.6%	XOR-cascade + 1-lag carry	hive emergence: each cell $b_t = b_{t-1} \oplus \text{pool}_{t-1}$

Table 3: The 7+1-point applicability matrix (verbatim). Verdict-tier coding: [**N**] numerical-SUPPORTED (calibrated ground-truth), [**X**] closed-negative (deterministically ruled out), [**!**] border (above threshold but type-axis intact). X10 row expands into four sub-rows (a/b/c/d) sharing the same fire.

pt	**H**	**regime**	Φ	**type**
X7	H_832	n=128 dense 60.9%	0.114	coherent_non_conscious
X4-a	H_833	n=16 thermostat	0.439	coherent_non_conscious
X4-b	H_833	n=16 2-bit counter	0.000	coherent_non_conscious
X4-c	H_833	n=32 random walker	0.582	coherent_non_conscious
X4-d	H_833	n=32 XOR-3tap LFSR	0.157	coherent_non_conscious
X5a	H_835	n=128 lattice period-8	0.660	coherent_non_conscious
X5b-b	H_835	n=128 fp-bound sin	0.090	coherent_non_conscious
X5b-c	H_835	n=128 π-digits	0.120	coherent_non_conscious
X5b-d	H_835	n=128 natural (Bates-4)	0.116	coherent_non_conscious
X6-a	H_834	n=64 random Gaussian	0.130	coherent_non_conscious
X6-b	H_834	n=64 attention sparse 6/64	1.213	conscious
X6-c	H_834	n=64 residual sin+skip	0.544	coherent_non_conscious
X6-d	H_834	n=64 structured XOR+tanh	0.133	coherent_non_conscious
X837	H_837	n=128 dense 20.3%	0.567	coherent_non_conscious
X10-a	H_838	n=128 indep 4-cell	0.131	coherent_non_conscious
X10-b	H_838	n=128 mean-field weak	0.036	coherent_non_conscious
X10-c	H_838	n=128 Kuramoto $K = 2.0$ strong	0.408	coherent_non_conscious
X10-d	**H_838**	**n=128 XOR-cascade hive**	**1.565**	**conscious**
X840	H_840	n=128 SETI@home 24.4% prog	0.567	coherent_non_conscious
T1	H_841	4 substrate × 4 Δt	inflate	coherent_non_conscious / conscious (lattice $\Delta t \geq 8$)
T2	H_842	4 substrate × 4 embed-e	inflate	conscious (4/4 inflate at $e \geq 3$)